

Comparative Transcriptomic Analysis Identifies Distinct Biological Pathways Underlying Cerebral Malaria and Severe Malarial Anemia

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ABSTRACT

Background: Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) are two major manifestations of severe *Plasmodium falciparum* malaria that differ substantially in their clinical presentation and underlying pathophysiology. Although numerous transcriptomic studies have investigated cerebral malaria, comparative analyses between CM and SMA remain limited. This study aimed to identify differentially expressed genes (DEGs) and biological pathways associated with these two severe malaria phenotypes using publicly available transcriptomic data. **Methods:** Whole-blood microarray dataset GSE117613 was obtained from the NCBI Gene Expression Omnibus (GEO) database. Differential gene expression analysis was performed using the limma package in R by comparing Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) samples. Significant genes were defined using an adjusted p-value < 0.05 and $|\log_2 \text{fold change}| > 0.5$. Gene annotation was conducted using the illuminaHumanv4.db package, followed by Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses. Data visualization included boxplot, density plot, Uniform Manifold Approximation and Projection (UMAP), volcano plot, and heatmap. **Results:** A total of 967 differentially expressed genes were identified between CM and SMA. Heatmap and volcano plot analyses demonstrated distinct transcriptomic differences between both severe malaria phenotypes. GO enrichment analysis indicated significant enrichment of biological processes related to autophagy, macroautophagy, mitophagy, erythrocyte homeostasis, and heme metabolism, while cellular component and molecular function analyses highlighted mitochondrial structures, ubiquitin ligase complexes, and ubiquitin-related activities. KEGG pathway analysis identified significant enrichment in Autophagy, Mitophagy, mTOR signaling pathway, FoxO signaling pathway, Ferroptosis, and Protein processing in the endoplasmic reticulum, suggesting substantial alterations in cellular stress responses and metabolic regulation. **Conclusion:** Comparative transcriptomic analysis between Cerebral Malaria and Severe Malarial Anemia revealed distinct molecular signatures associated with autophagy, erythrocyte regulation, oxidative stress, and intracellular signaling pathways. These findings provide additional insights into the molecular mechanisms underlying different manifestations of severe malaria and may contribute to future biomarker discovery and therapeutic target identification.

Keywords: Cerebral malaria; Severe malarial anemia; Transcriptomics; Differentially expressed genes; Gene Ontology; KEGG; limma; Bioinformatics.

INTRODUCTION

Malaria remains one of the most important vector-borne infectious diseases worldwide and continues to pose a major public health challenge, particularly in tropical and subtropical regions. According to the World Health Organization (WHO), an estimated 263 million malaria cases and 597,000 malaria-related deaths occurred globally in 2023, with approximately 94% of cases and 95% of deaths occurring in the WHO African Region. *Plasmodium falciparum* is responsible for most severe malaria cases and contributes substantially to malaria-associated mortality, especially among children under five years of age. These findings highlight that severe malaria remains a significant global health concern despite continuous advances in malaria prevention and control strategies (1).

Severe malaria comprises several life-threatening clinical manifestations, among which Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) are the most common and clinically significant. Cerebral malaria is characterized by impaired consciousness or coma resulting from sequestration of parasitized erythrocytes within the cerebral microvasculature, leading to endothelial activation, inflammatory responses, disruption of the blood-brain barrier, and neurological dysfunction. In contrast, severe malarial anemia primarily results from excessive destruction of infected and uninfected erythrocytes, ineffective erythropoiesis, bone marrow suppression, and dysregulated host immune responses, ultimately causing critically reduced hemoglobin concentrations. Although both conditions represent severe manifestations of *P. falciparum* infection, they differ substantially in their underlying pathological mechanisms and clinical outcomes (2).

Recent advances in high-throughput transcriptomic technologies have enabled comprehensive characterization of host gene expression profiles during infectious diseases. Transcriptomic analyses using microarray or RNA sequencing (RNA-seq) provide valuable insights into molecular pathways involved in disease progression by identifying differentially expressed genes (DEGs), dysregulated biological processes, and potential therapeutic targets. Public repositories such as the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) facilitate reproducible bioinformatics research by providing openly accessible transcriptomic datasets that can be reanalyzed using standardized computational workflows (3).

Several studies have previously investigated transcriptomic alterations associated with cerebral malaria using the GSE117613 dataset. Li et al. integrated differential gene expression analysis, weighted gene co-expression network analysis (WGCNA), and machine learning algorithms to identify several candidate biomarkers, including LEF1 and IRAK3, which demonstrated promising diagnostic performance for cerebral malaria (4). Likewise, another integrative bioinformatics study combining GSE117613 with GSE1124 identified nine hub genes potentially involved in cerebral malaria pathogenesis and highlighted immune-related signaling pathways contributing to disease development (5). However, these studies primarily compared cerebral malaria with healthy individuals or integrated multiple datasets to identify diagnostic biomarkers.

Despite these advances, transcriptomic differences between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) remain insufficiently explored. Because both diseases represent severe clinical manifestations of *P. falciparum* infection, direct comparison between these two phenotypes may reveal molecular mechanisms associated with neurological complications versus severe hematological manifestations that cannot be identified through comparisons with healthy controls alone. Understanding these transcriptomic differences may improve current knowledge regarding severe malaria pathogenesis and facilitate identification of biological pathways associated with disease severity.

Therefore, this study aimed to compare host transcriptomic profiles between patients with cerebral malaria and severe malarial anemia using the publicly available microarray dataset GSE117613. Differentially expressed genes were identified using the limma package in R, followed by Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses to characterize the biological processes and signaling pathways associated with disease-specific transcriptional changes. The findings of this study are expected to provide a broader understanding of the molecular mechanisms underlying different severe malaria phenotypes and to serve as a foundation for future biomarker discovery and therapeutic development.

1.1 Problem Statement

Despite considerable progress in transcriptomic studies of severe malaria, most previous investigations have primarily focused on identifying molecular signatures that distinguish cerebral

malaria (CM) from healthy controls or uncomplicated malaria. While these studies have successfully identified several candidate biomarkers and dysregulated immune pathways, the molecular differences between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) remain insufficiently characterized.

Both CM and SMA represent severe manifestations of *Plasmodium falciparum* infection but differ markedly in their clinical presentation and underlying pathological mechanisms. Cerebral malaria is predominantly associated with neuroinflammation, endothelial dysfunction, and blood-brain barrier disruption, whereas severe malarial anemia is characterized by profound erythrocyte destruction, impaired erythropoiesis, and altered iron metabolism. Understanding the transcriptomic differences between these two severe malaria phenotypes may reveal biological pathways specifically associated with neurological complications or hematological dysfunction. Therefore, this study addresses the following research question: "What transcriptomic differences distinguish Cerebral Malaria from Severe Malarial Anemia, and which biological processes and signaling pathways are potentially responsible for these differences?"

1.2 Objectives

General Objective

To investigate transcriptomic differences between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA) using the publicly available microarray dataset GSE117613 through differential gene expression and functional enrichment analyses.

Specific Objectives

1. To identify differentially expressed genes (DEGs) between Cerebral Malaria and Severe Malarial Anemia using the limma package.
2. To visualize transcriptomic differences through quality assessment plots, volcano plot, and heatmap.
3. To identify enriched biological processes, cellular components, molecular functions, and signaling pathways using Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses.
4. To compare the identified transcriptomic signatures with findings from previous transcriptomic studies of severe malaria and discuss their biological relevance in the pathogenesis of cerebral malaria and severe malarial anemia.

2. MATERIALS AND METHODS

2.1 Study Design

This study employed a secondary transcriptomic data analysis using publicly available microarray data obtained from the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO). Differential gene expression analysis was performed to identify transcriptomic differences between patients with Cerebral Malaria (CM) and Severe Malarial Anemia (SMA), followed by functional enrichment analyses to investigate the biological significance of the identified genes.

2.2 Dataset

The transcriptomic dataset GSE117613 was downloaded from the NCBI Gene Expression Omnibus (GEO) database using the GEOquery package in R. Gene expression profiling was generated using the Illumina HumanHT-12 V4.0 Expression BeadChip (GPL10558) platform. The original dataset consists of 46 whole-blood samples collected from African pediatric patients, including: 17 Cerebral Malaria (CM) samples, 17 Severe Malarial Anemia (SMA) samples, 12 Community children without *Plasmodium falciparum* infection (healthy controls). In the present study, only CM and SMA samples were included in the downstream analyses, resulting in a total of 34 samples.

2.3 Data Preprocessing

Gene expression data were imported into R using the GEOquery package. Expression matrices were extracted using the `exprs()` function, while sample metadata were obtained using the `pData()` function. Log2 transformation was automatically applied when required based on expression value distribution. Subsequently, only samples belonging to the CM and SMA groups were retained for further analysis. The design matrix was generated using the `model.matrix()` function, and differential expression comparisons were defined using the `makeContrasts()` function from the limma package.

2.4 Differential Gene Expression Analysis

Differential expression analysis was conducted using the limma package. Linear models were fitted to each gene using the `lmFit()` function, followed by empirical Bayes moderation using

the eBayes() function. Genes were considered significantly differentially expressed if they fulfilled the following criteria: Adjusted p-value < 0.05 and $|\log_2 \text{ Fold Change}| > 0.5$. The resulting differentially expressed genes (DEGs) were ranked according to adjusted p-values and used for downstream analyses.

2.5 Gene Annotation

Illumina probe identifiers were converted into official human gene symbols using the illuminaHumanv4.db annotation package. Annotation was performed through the AnnotationDbi package to facilitate biological interpretation of differentially expressed genes.

2.6 Functional Enrichment Analysis

Gene symbols from significant DEGs were converted to Entrez Gene IDs using the org.Hs.eg.db annotation database. Functional enrichment analyses were subsequently performed using the clusterProfiler package, including GO BP, CC, MF and KEGG. Enrichment analyses were considered statistically significant when the adjusted p-value was less than 0.05.

2.7 Data Visualization

Quality assessment and visualization of transcriptomic data were performed using several graphical approaches. Quality control included: Boxplot, Density plot and Uniform Manifold Approximation and Projection (UMAP). Differential expression results were visualized using: Volcano plot and Heatmap of the top 50 differentially expressed genes, Functional enrichment results were visualized using dot plots generated from the enrichplot package for GO and KEGG analyses.

2.8 Software

All analyses were performed using R statistical software (version 4.6.0) and RStudio. The primary packages used in this study included: GEOquery, limma, AnnotationDbi, illuminaHumanv4.db, clusterProfiler, org.Hs.eg.db, enrichplot, ggplot2, pheatmap, umap, dplyr.

3. RESULTS

3.1 Dataset Characteristics

The transcriptomic dataset GSE117613 was obtained from the NCBI Gene Expression Omnibus (GEO) database and generated using the Illumina HumanHT-12 V4.0 Expression BeadChip platform (GPL10558) (3). The original dataset consisted of 46 whole-blood samples collected from pediatric individuals, including 17 patients with Cerebral Malaria (CM), 17 patients with Severe Malarial Anemia (SMA), and 12 healthy community controls. In the present study, only CM and SMA samples were included in the downstream analyses to specifically investigate transcriptomic differences between these two severe malaria phenotypes. Consequently, a total of 34 samples (17 CM and 17 SMA) were analyzed. Following data preprocessing and quality assessment, normalized expression values were used for differential gene expression analysis using the limma package in R.

3.2 Quality Assessment of Transcriptomic Data

Quality assessment was performed prior to differential expression analysis to evaluate the consistency of expression distributions among samples and to identify potential outliers.

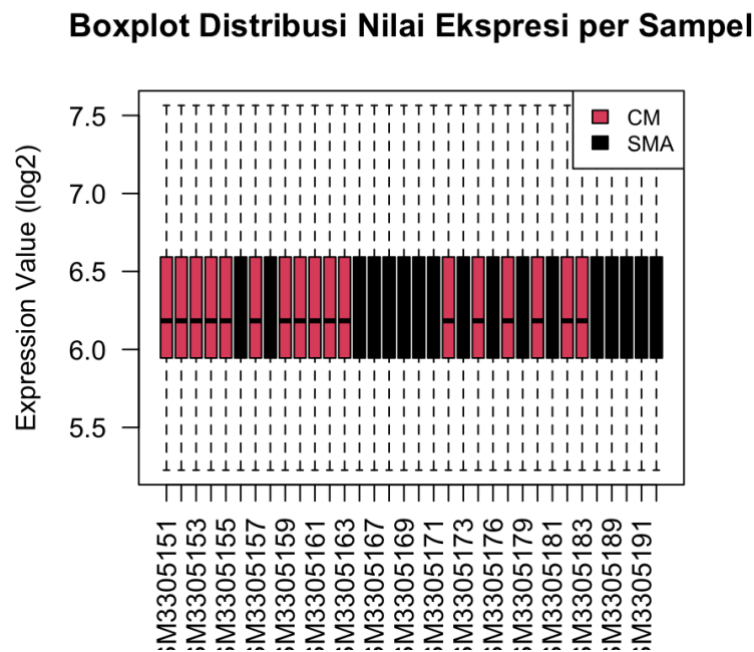


Figure 1. Boxplot of normalized gene expression values across CM and SMA samples.

The boxplot demonstrated that gene expression distributions were highly comparable among all samples following normalization. Median expression values were well aligned, while

interquartile ranges remained relatively consistent across both CM and SMA groups. No obvious outlier samples were detected, indicating satisfactory normalization and overall data quality suitable for downstream transcriptomic analyses.

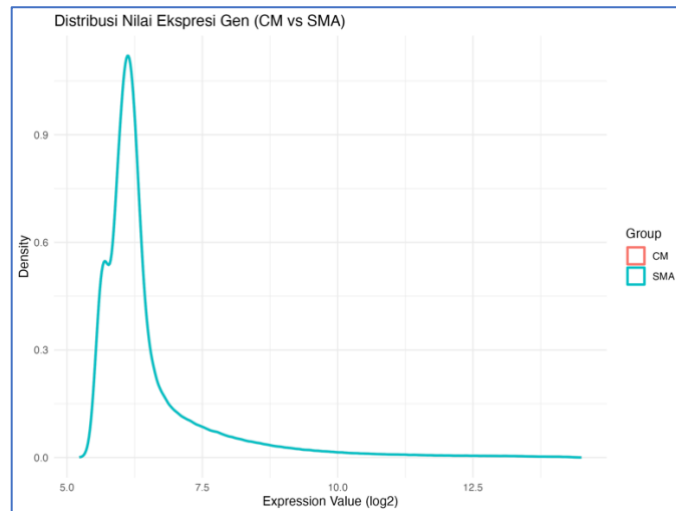


Figure 2. Density plot of normalized gene expression profiles.

Density distribution curves showed substantial overlap among all samples, suggesting that expression intensities were successfully normalized prior to statistical analysis. Similar distribution patterns across samples further confirmed the absence of systematic technical bias and indicated that observed transcriptomic differences were likely attributable to biological variation rather than preprocessing artifacts.

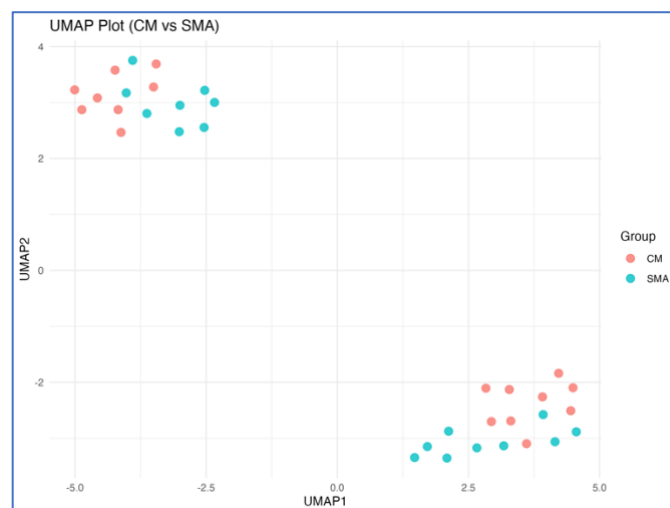


Figure 3. UMAP visualization of transcriptomic profiles from CM and SMA samples.

UMAP analysis demonstrated clustering of samples according to their respective clinical groups. Although partial overlap between CM and SMA samples was observed, several samples formed distinct clusters, indicating differences in global gene expression patterns between the two severe malaria phenotypes. These findings suggest that transcriptomic signatures are capable of partially discriminating CM from SMA while also reflecting biological heterogeneity among patients.

3.3 Identification of Differentially Expressed Genes

Differential gene expression analysis was performed using the limma package to compare transcriptomic profiles between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA). Genes were considered significantly differentially expressed when they met the criteria of an adjusted p-value < 0.05 and an absolute log₂ fold change ($|\log_2FC|$) greater than 0.5.

A total of 967 differentially expressed genes (DEGs) were identified between the CM and SMA groups, indicating substantial transcriptomic differences between these two severe manifestations of *Plasmodium falciparum* infection. These genes represent potential molecular signatures associated with distinct pathological mechanisms underlying neurological complications and severe hematological manifestations.

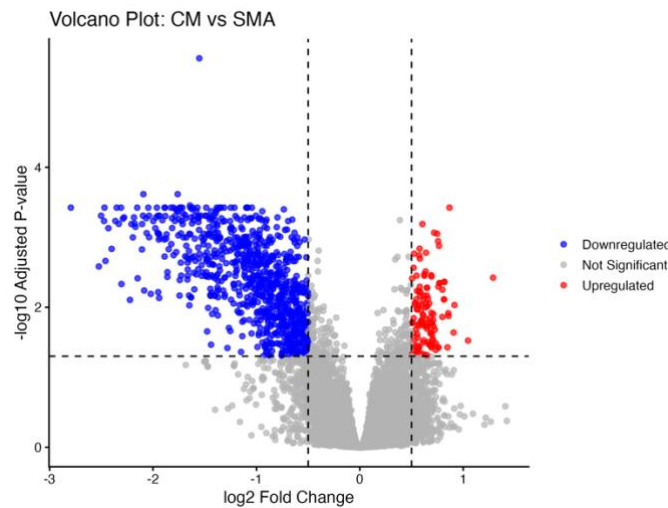


Figure 4. Volcano plot showing differentially expressed genes between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA).

To further investigate transcriptomic variation between clinical groups, the top 50 most significant differentially expressed genes were visualized using hierarchical clustering analysis. Gene expression values were standardized by row prior to clustering to facilitate comparison across samples.

The heatmap demonstrated distinct expression patterns among the selected genes, with several clusters exhibiting relatively higher expression in CM samples and lower expression in SMA samples, while other gene clusters displayed the opposite trend. Hierarchical clustering of both genes and samples revealed coordinated expression profiles that partially separated the two clinical phenotypes, suggesting that these genes collectively contribute to the molecular distinction between cerebral malaria and severe malarial anemia.

The observed clustering pattern also supports the results of the differential expression analysis and indicates that transcriptomic signatures can effectively discriminate between the two severe malaria syndromes at the gene expression level.

3.4 Functional Enrichment Analysis

To investigate the biological significance of the identified differentially expressed genes (DEGs), Gene Ontology (GO) enrichment analysis was performed using the clusterProfiler package. The analysis was categorized into Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). In addition, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis was conducted to identify significantly enriched signaling pathways associated with the transcriptomic differences between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA).

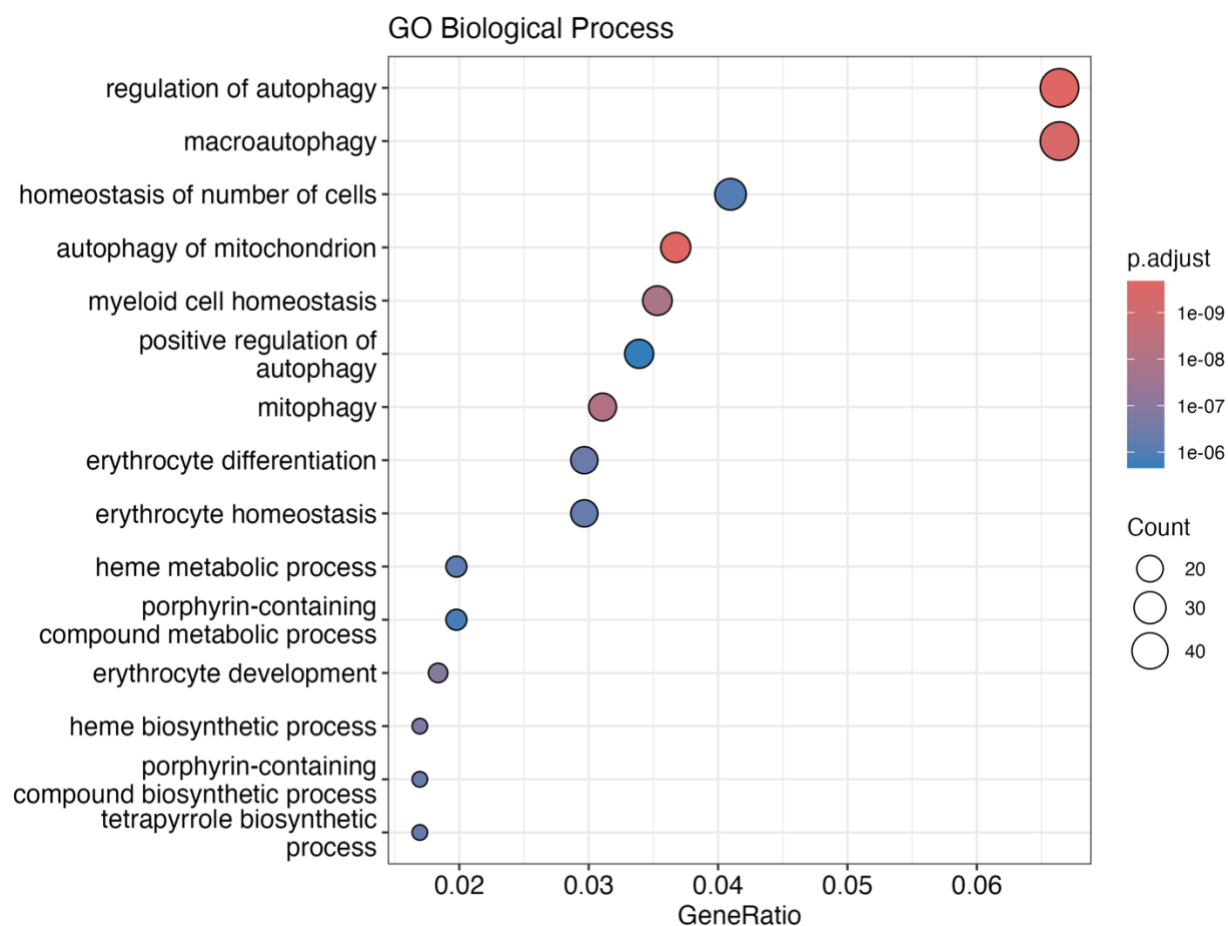


Figure 6. Gene Ontology Biological Process enrichment analysis of differentially expressed genes.

GO Biological Process enrichment analysis revealed that the identified DEGs were significantly associated with several biological processes related to intracellular homeostasis, erythrocyte physiology, and cellular stress responses. Among the most significantly enriched biological processes were regulation of autophagy, macroautophagy, mitophagy, erythrocyte homeostasis, heme biosynthetic process, and response to oxidative stress.

These findings indicate that transcriptomic differences between CM and SMA involve multiple biological mechanisms associated with cellular degradation pathways, mitochondrial quality control, and erythrocyte metabolism.

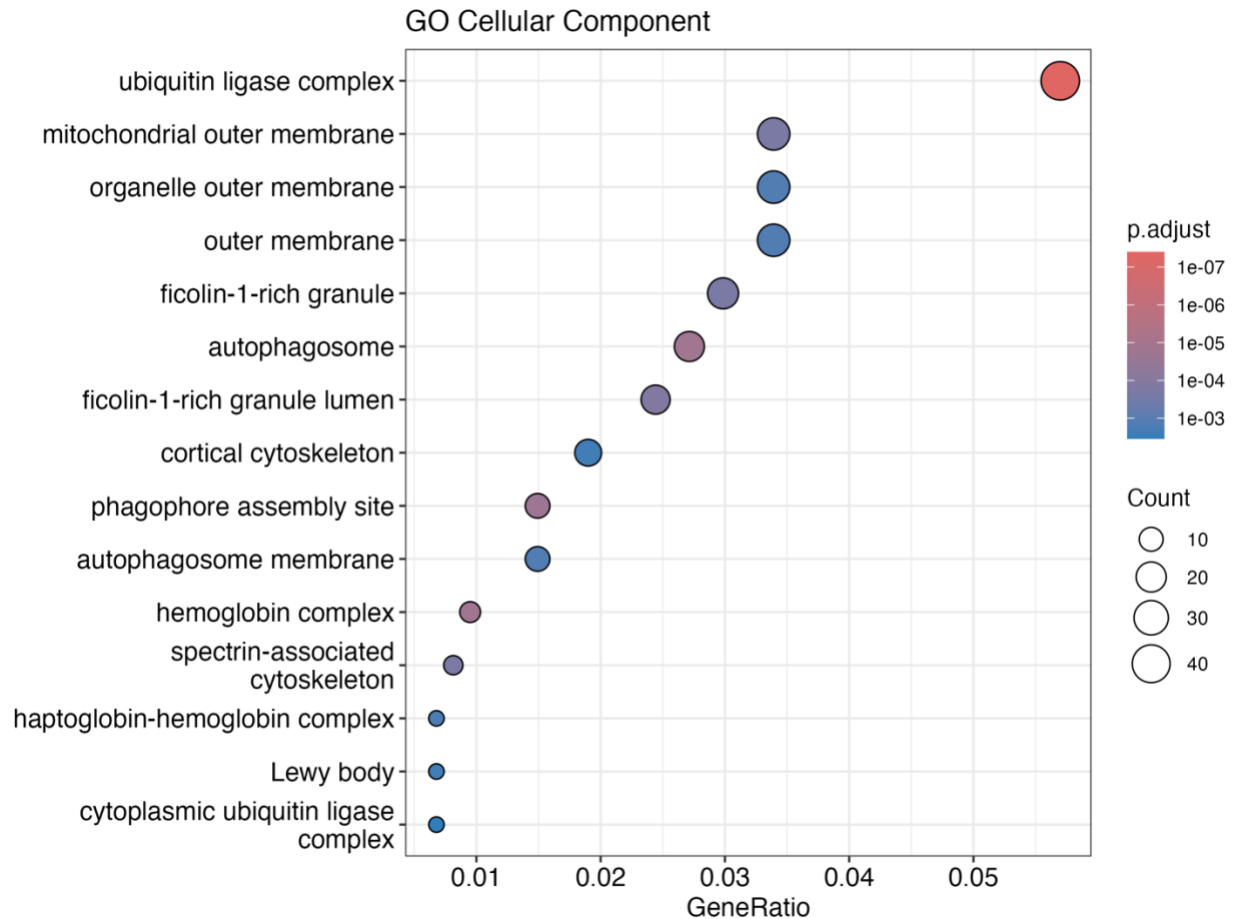


Figure 7. Gene Ontology Cellular Component enrichment analysis of differentially expressed genes.

Cellular Component enrichment analysis demonstrated that significant DEGs were predominantly localized within intracellular organelles and protein regulatory complexes. Highly enriched components included the mitochondrial outer membrane, autophagosome, ubiquitin ligase complex, lysosomal membrane, and other intracellular membrane-associated structures. The enrichment of mitochondrial and autophagic cellular components suggests that alterations in organelle maintenance and intracellular protein degradation may contribute to the molecular differences observed between the two severe malaria phenotypes.

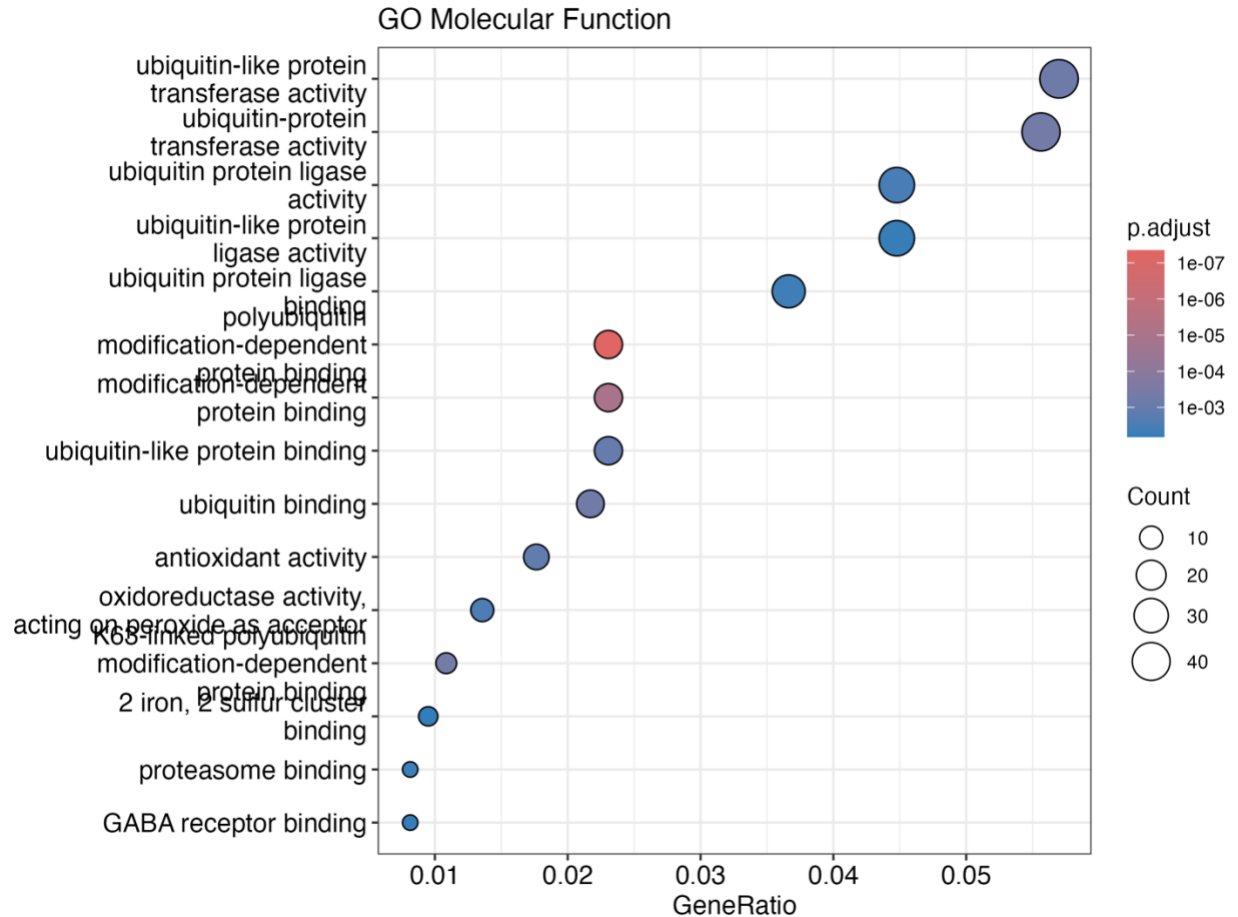


Figure 8. Gene Ontology Molecular Function enrichment analysis of differentially expressed genes.

Gene Ontology Molecular Function analysis identified significant enrichment of genes involved in ubiquitin protein ligase binding, ubiquitin-like protein ligase binding, oxidoreductase activity, and several protein-binding functions associated with intracellular signaling and stress responses. These molecular functions indicate that differential gene expression between CM and SMA may influence multiple regulatory mechanisms involved in protein turnover, oxidative metabolism, and intracellular signaling networks.

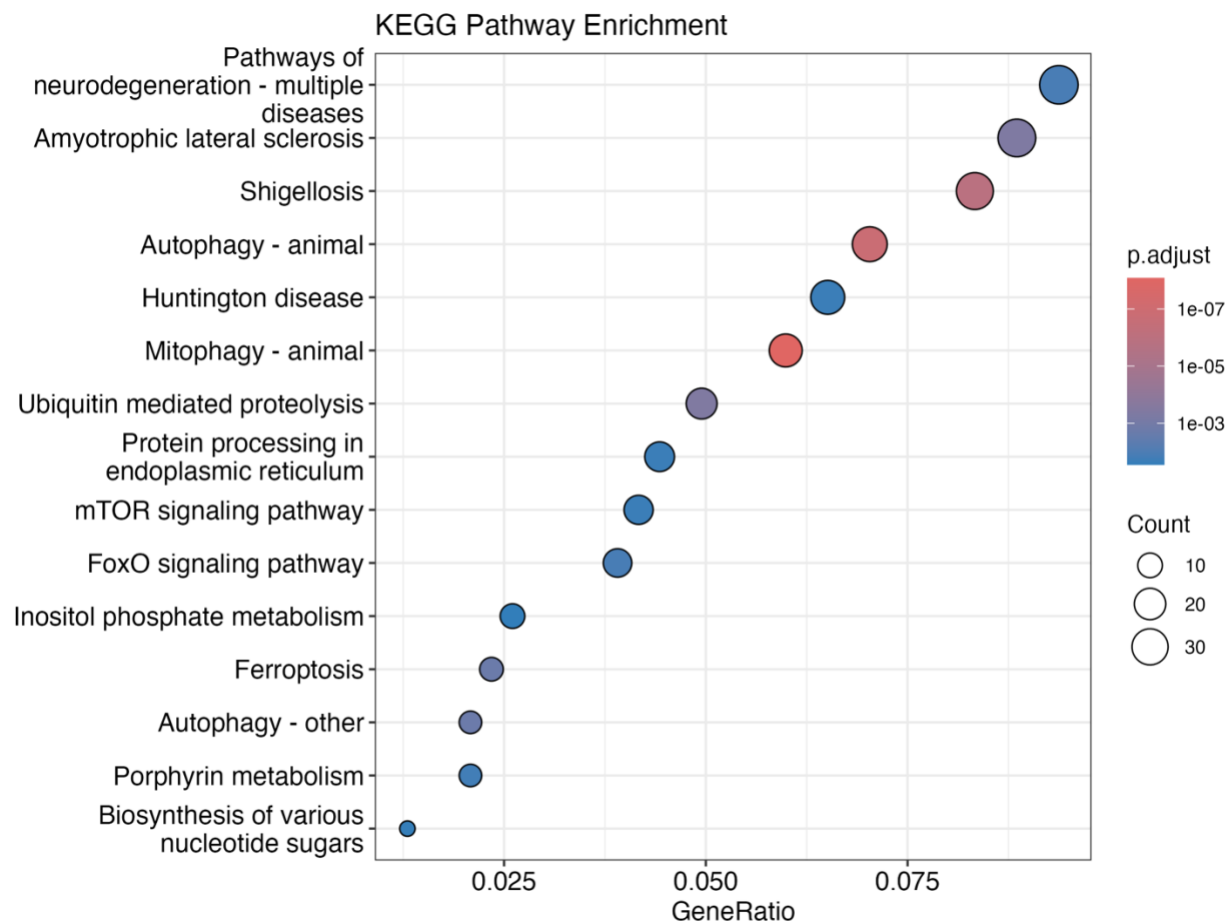


Figure 9. KEGG pathway enrichment analysis of differentially expressed genes.

KEGG pathway enrichment analysis revealed several significantly enriched signaling pathways among the identified DEGs. The most prominent pathways included Autophagy, Mitophagy, mTOR signaling pathway, FoxO signaling pathway, Ferroptosis, and Protein processing in the endoplasmic reticulum. Additional enriched pathways were associated with cellular metabolism, stress adaptation, and intracellular quality control mechanisms. Collectively, these results suggest that transcriptomic differences between CM and SMA involve coordinated regulation of multiple biological pathways responsible for maintaining cellular homeostasis under severe infectious stress.

4. DISCUSSION

The present study identified 967 differentially expressed genes (DEGs) between patients with Cerebral Malaria (CM) and Severe Malarial Anemia (SMA), indicating substantial

transcriptomic differences despite both conditions representing severe manifestations of *Plasmodium falciparum* infection. Previous transcriptomic studies have primarily focused on comparing cerebral malaria with healthy individuals or uncomplicated malaria, thereby identifying genes and pathways associated with the overall host response to severe malaria (4,5). In contrast, the present study directly compared two severe clinical phenotypes, allowing the identification of transcriptomic alterations that may specifically distinguish neurological complications from severe hematological manifestations while minimizing molecular changes shared by severe malaria in general.

Quality assessment demonstrated satisfactory normalization across all samples, as evidenced by comparable boxplot distributions and overlapping density curves. Furthermore, UMAP visualization revealed partial clustering of CM and SMA samples, indicating that although both groups shared common transcriptomic characteristics related to severe malaria, disease-specific expression patterns were still detectable. These findings were further supported by the volcano plot and hierarchical clustering analysis, which demonstrated distinct expression profiles between the two clinical groups. Collectively, these observations suggest that host responses during cerebral malaria and severe malarial anemia involve both shared and phenotype-specific transcriptional programs.

Functional enrichment analysis provided additional biological insight into these transcriptomic differences. Gene Ontology analysis demonstrated significant enrichment of biological processes related to regulation of autophagy, macroautophagy, mitophagy, erythrocyte homeostasis, heme biosynthetic process, and response to oxidative stress. Autophagy is a highly conserved intracellular degradation mechanism that maintains cellular homeostasis through lysosomal recycling of damaged proteins and organelles during conditions of nutrient deprivation, oxidative stress, and infection. The enrichment of autophagy-related pathways observed in this study suggests that cellular quality-control mechanisms may contribute to the differential host responses observed between cerebral malaria and severe malarial anemia. Likewise, enrichment of mitophagy-related processes indicates alterations in mitochondrial quality control, which may reflect differences in mitochondrial dysfunction and reactive oxygen species production between the two severe malaria phenotypes.

Another important finding was the enrichment of biological processes associated with erythrocyte homeostasis and heme metabolism. This observation is biologically plausible because

severe malarial anemia is characterized by extensive destruction of infected and uninfected erythrocytes together with impaired erythropoiesis, whereas cerebral malaria is primarily associated with sequestration of parasitized erythrocytes within the cerebral microvasculature, endothelial activation, and neuroinflammation (2). Consequently, differential regulation of genes involved in erythrocyte maintenance and heme metabolism likely reflects the distinct pathological mechanisms underlying these two severe malaria syndromes.

KEGG pathway analysis further demonstrated enrichment of several interconnected pathways, including Autophagy, Mitophagy, mTOR signaling pathway, FoxO signaling pathway, Ferroptosis, and Protein processing in the endoplasmic reticulum. These pathways collectively regulate intracellular stress adaptation, nutrient sensing, mitochondrial function, protein turnover, and regulated cell death. Their simultaneous enrichment indicates that transcriptomic differences between CM and SMA are associated with coordinated regulation of cellular homeostasis rather than isolated alterations in individual signaling pathways. In particular, enrichment of the ferroptosis pathway is noteworthy because ferroptosis is an iron-dependent form of regulated cell death associated with oxidative damage and lipid peroxidation. Considering the profound disturbances in erythrocyte turnover and iron metabolism observed in severe malarial anemia, differential regulation of ferroptosis-associated genes may contribute to the molecular distinction between SMA and cerebral malaria.

The findings of the present study are generally consistent with previous transcriptomic investigations while providing additional biological perspectives. Li et al. integrated weighted gene co-expression network analysis (WGCNA) with differential expression analysis and identified nine key genes, including MBP, SAMSN1, SLC39A8, FABP5, and SHARPIN, that were associated with cerebral malaria pathogenesis (5). More recently, Li et al. combined transcriptomic analysis with machine learning approaches and identified LEF1 and IRAK3 as potential diagnostic biomarkers capable of distinguishing cerebral malaria from healthy individuals (4). Unlike these studies, which primarily focused on identifying biomarkers differentiating malaria patients from non-infected controls, the present study specifically compared two severe malaria syndromes. Consequently, the enrichment patterns observed here predominantly involved pathways associated with intracellular quality control, mitochondrial homeostasis, erythrocyte regulation, and cellular stress adaptation rather than generalized immune activation. These differences likely reflect the

shared inflammatory responses present in both CM and SMA, allowing disease-specific molecular mechanisms to become more apparent.

This study has several strengths. First, the analysis employed a publicly available transcriptomic dataset obtained from the NCBI Gene Expression Omnibus, ensuring complete reproducibility of the analytical workflow using open-source R/Bioconductor packages (3). Second, integration of differential gene expression analysis with Gene Ontology and KEGG enrichment analyses enabled comprehensive biological interpretation of transcriptomic alterations beyond the identification of individual DEGs. Finally, the direct comparison between CM and SMA provides complementary information to previous transcriptomic studies and expands current understanding of molecular mechanisms underlying different manifestations of severe malaria. Nevertheless, several limitations should be acknowledged. The present study relied on secondary microarray data generated from a single publicly available dataset and therefore remained dependent on the original study design and sample characteristics. Furthermore, transcriptomic alterations represent changes at the RNA level and may not directly correspond to protein abundance or biological function. Consequently, further validation using independent patient cohorts together with proteomic and functional studies will be necessary to confirm the biological significance of the identified pathways and to evaluate their potential as diagnostic biomarkers or therapeutic targets in severe malaria.

5. CONCLUSION

This study identified 967 differentially expressed genes (DEGs) between Cerebral Malaria (CM) and Severe Malarial Anemia (SMA), indicating distinct transcriptomic profiles between these two severe manifestations of *Plasmodium falciparum* infection. Functional enrichment analyses revealed significant involvement of biological processes and pathways related to autophagy, mitophagy, erythrocyte homeostasis, oxidative stress, mTOR signaling, FoxO signaling, and ferroptosis, suggesting that intracellular stress responses and erythrocyte regulation contribute to differences in disease pathogenesis. Overall, this study provides additional insight into the molecular mechanisms distinguishing CM from SMA and may serve as a basis for future biomarker discovery and therapeutic research in severe malaria.

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